THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS · MEDIUM LEVEL CHAPTER TEST SET 2

Chapter 3: **A Story of Numbers** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The largest 3-digit number in base 3, written in decimal, is: 1

(a) 27 **(b)** 8 **(c)** 26 **(d)** 25

2. The number of digits in the binary form of 50 is: 1

(a) 6 **(b)** 8 **(c)** 5 **(d)** 7

3. $1101_2 + 101_2$ equals: 1

(a) 10001_2 **(b)** 10100_2 **(c)** 10010_2 **(d)** 10011_2

4. In 3721_8, the value (in decimal) contributed by the digit 3 in position 1 from the left is: 1

(a) 24 **(b)** 1536 **(c)** 515 **(d)** 12288

5. The decimal value of 2102_3 is: 1

(a) 67 **(b)** 71 **(c)** 65 **(d)** 63

6. The binary form of 37 is: 1

(a) 100110 **(b)** 100101 **(c)** 100100 **(d)** 101001

7. XCVI + CCVIII equals: 1

(a) CCXCIV **(b)** CCCXIV **(c)** CCCV **(d)** CCCIV

8. The largest 3-digit number in base 5, written in decimal, is: 1

(a) 14 **(b)** 125 **(c)** 124 **(d)** 24

9. The number of digits in the binary form of 127 is: 1

(a) 8 **(b)** 7 **(c)** 6 **(d)** 9

10. $1001_2 + 111_2$ equals: 1

(a) 1111_2 **(b)** 10001_2 **(c)** 10000_2 **(d)** 10010_2

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Write LXXXVII + DXIII as a Roman numeral. 2

12. Subtract $100011_2 - 1100_2$ and give the answer in binary. 2

13. Convert 3402_5 to decimal. 2

14. Write CLXXVI + CDXXIV as a Roman numeral. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Convert 571_8 to base 5 by first converting to decimal. 3

16. Multiply $111_2 \times 11_2$ and give the answer in binary. 3

17. Write 1486 and 349 in Roman numerals and find their difference in Roman numerals. 3

OR

Convert 2201_3 to base 5 by first converting to decimal. 3

18. Multiply $1001_2 \times 101_2$ and give the answer in binary. 3

OR

Write 3000 and 1776 in Roman numerals and find their difference in Roman numerals. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Find the sum of 10110_2, 111_2 and 1100_2. Verify the result in decimal. 5

OR

Write the smallest and the largest 3-digit numbers in base 5 and find their decimal values. How many 3-digit numbers are there in base 5? 5

20. Find the sum of 11001_2, 1110_2 and 1001_2. Verify the result in decimal. 5

OR

Write the smallest and the largest 4-digit numbers in base 4 and find their decimal values. How many 4-digit numbers are there in base 4? 5

END OF QUESTION PAPER — Total: 40 Marks

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